

Fever in Infants and Children

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Reviewed/Revised Mar 2025 | Modified May 2026

Normal body temperature varies from person to person and throughout the day (it is typically highest in the afternoon). Normal body temperature is higher in preschool-aged children and highest at approximately 18 to 24 months of age. However, despite these variations, most doctors define fever as a temperature of 100.4° F (approximately 38° C) or higher when measured with a rectal thermometer (see [How to Take A Child's Temperature](#)).

Although parents often worry about how high the temperature is, the height of the fever does not necessarily indicate how serious the cause is. Some minor illnesses cause a high fever, and some serious illnesses cause only a mild fever. Other symptoms (such as difficulty breathing, confusion, and not drinking) indicate the severity of illness much better than the temperature does. However, a temperature over 106° F (approximately 41° C), although quite rare, can itself be dangerous.

Fever can be useful in helping the body fight infection. Some experts think that reducing fever can make some illnesses last longer or possibly interfere with the immune system's response to infection. Thus, although a fever is uncomfortable, it does not always require treatment in otherwise healthy children. However, in children with a lung, heart, or brain disorder, fever may cause problems because it increases demands on the body (for example, by increasing the heart rate). So lowering the temperature in such children is important.

Infants with a fever are usually irritable and may not sleep or feed well. Older children lose their interest in play. Usually, the higher a fever gets, the more irritable and disinterested children become. However, sometimes children with a high fever look surprisingly well. Children may have seizures when their temperature rises or falls rapidly (called [febrile seizures](#)). Rarely, a fever gets so high that children become listless, drowsy, and unresponsive.

(See also [Fever in Adults](#).)

How to Take a Child's Temperature



A child's temperature can be taken from the rectum, ear, mouth, forehead, or armpit. It can be taken with a digital thermometer. Digital thermometers are easier to use and give much quicker readings (and usually give a signal when they are ready). Glass thermometers containing mercury are not recommended because they can break and expose people to mercury.

Rectal temperatures are most accurate. That is, they come closest to the child's true internal body temperature. For a rectal temperature, the bulb of the thermometer should be coated with a lubricant. Then the thermometer is gently inserted approximately 1/2 to 1 inch (approximately 1 1/4 to 2 1/2 centimeters) into the rectum while the child is lying face down. The child should be kept from moving.

Oral temperatures are taken by placing a digital thermometer under the child's tongue. Oral temperatures provide reliable readings but are difficult to take in young children. Young children have difficulty keeping their mouth gently closed around the thermometer, which is necessary for an accurate reading. The age at which oral temperatures can be reliably taken varies from child to child but is typically after age 4.

Armpit temperatures are taken by placing a digital thermometer in the child's armpit, directly on the skin. Doctors rarely use this method because it is less accurate than others (readings are usually too low and vary greatly). However, if caretakers are uncomfortable taking a rectal temperature and do not have a device to measure ear or forehead temperature, measuring armpit temperature may be better than not measuring temperature at all.

Ear temperatures are taken with a digital device that measures infrared (heat) radiation from the eardrum. Ear thermometers are unreliable in infants under 3 months old. For an ear temperature, the thermometer probe is placed around the opening of the ear so that a seal is formed, then the start button is pressed. A digital readout provides the temperature.

Forehead temperatures (temporal artery temperatures) are taken with a digital device that measures infrared radiation from an artery in the forehead (the temporal artery). For a forehead temperature, the head of the thermometer is moved lightly across the forehead from hairline to hairline while pressing the

scan button. A digital readout provides the temperature. Forehead temperatures are not as accurate as rectal temperatures, particularly in infants under 3 months old.

How to Take the Temperature in an Infant or a Child



Causes of Fever in Infants and Children

Fever occurs in response to infection, injury, or inflammation and has many causes. Likely causes of fever depend on whether it has lasted 14 days or less (acute) or more than 14 days (chronic), as well as on the age of the child. Fevers are usually acute.

Acute fever

Acute fevers in infants and children are usually caused by an infection.

The **most common causes of acute fever (lasting 14 days or less)** are

- Respiratory infections due to a virus, such as colds or flu
- [Gastroenteritis](#) (infection of the digestive tract) due to a virus
- Certain bacterial infections, particularly [ear infections](#) (otitis media), sinus infections, [pneumonia](#), and [urinary tract infections](#)

Newborns and young infants are at higher risk of certain serious infections because their immune system is not fully developed. Such infections may be acquired before birth or during birth and include [sepsis](#) (a serious infection of the blood), pneumonia (infection of the small air sacs of the lungs), and [meningitis](#) (infection of the tissues covering the brain).

Children under 3 years old who develop a fever (particularly if their temperature is 102.2° F [39° C] or higher) sometimes have bacteria in their bloodstream ([bacteremia](#)). Unlike older children, they sometimes have bacteremia with no symptoms besides fever (called [occult bacteremia](#)). [Routine vaccines](#) against the bacteria that usually cause occult bacteremia ([Streptococcus pneumoniae](#) and [Haemophilus influenzae type b \[Hib\]](#)) are now widely used in the United States and Europe. As a result, these vaccines have nearly eliminated occult bacteremia in children in this age group.

Less common causes of acute fever include side effects of vaccinations and of certain medications, bacterial infections of the skin ([cellulitis](#)) or joints ([septic arthritis](#)), viral or bacterial infections of the brain ([encephalitis](#)) or the tissues covering the brain ([meningitis](#)), or disorders where different parts of the body become inflamed ([Kawasaki disease](#) or [multisystem inflammatory syndrome in children](#) [MIS-C]). [Heatstroke](#) causes a very high body temperature.

Typically, a fever due to vaccination lasts a few hours to a day after the vaccine is given. However, some vaccinations can cause a fever even 1 or 2 weeks after the vaccine is given (as with [measles vaccination](#)). Children who have a fever when they are scheduled to receive a vaccine can still receive the vaccine if the fever is low and they have no serious illness.

[Teething](#) does not typically cause fever over 101° F (38.3° C).

Chronic fever

The **most common causes of chronic fever (lasting 14 days or more)** are

- A prolonged viral illness
- Back-to-back viral illnesses, especially in young children

Chronic fever, however, can also be caused by many other infectious and noninfectious disorders.

Infectious causes of chronic fever include

- [Hepatitis](#)
- [Sinusitis](#)
- [Pneumonia](#)
- [Abdominal abscesses](#) (pockets of pus in the abdomen)
- Infections of the digestive tract caused by bacteria or parasites
- Bone infections (such as [osteomyelitis](#))
- Heart infections (such as [endocarditis](#))

- [Tuberculosis](#)

Noninfectious causes of chronic fever include

- [Inflammatory bowel disease](#)
- [Juvenile idiopathic arthritis](#) or other [systemic rheumatic disorders](#)
- Cancer (such as [leukemia](#) and [lymphoma](#))

Occasionally, children fake a fever, or [caregivers fake a fever in the child they care for](#).

Sometimes the cause of a fever is not identified.

Evaluation of Fever in Infants and Children

Detecting a fever is not difficult, but determining its cause can be.

Warning signs

Certain symptoms are cause for concern. They include

- Any fever in infants less than 3 months old
- Lethargy or listlessness
- Ill appearance
- Difficulty breathing
- Bleeding in the skin, appearing as tiny reddish purple dots (petechiae) or splotches (purpura)
- Continuous crying in an infant or toddler (inconsolability)
- Headache, neck stiffness, confusion, or a combination in an older child

When to see a doctor

Children with fever should be evaluated by a doctor right away if they have any warning signs or are less than 3 months old.

Children who do not have warning signs but who are between 3 months and 3 years of age should be seen by the doctor if the fever is 102.2° F (39° C) or higher, if there are no obvious symptoms of an upper respiratory infection (for example, sneezing, a runny nose, or nasal congestion), or if the fever has continued for more than 1 or 2 days.

For children who do not have warning signs who are over 3 years of age, the need for and timing of a doctor's evaluation depend on the child's symptoms. Children who

have upper respiratory symptoms but otherwise appear well may not need further evaluation. Children over 3 years of age with fever lasting more than 5 days should be seen by the doctor even if they appear well.

What the doctor does

Doctors first ask questions about the child's symptoms and medical history. Doctors then do a physical examination. A description of the child's symptoms and a thorough examination usually enable doctors to identify the fever's cause (see table

Some Common Causes and Features of Fever in Children

Doctors take the child's temperature. It is measured rectally in infants and young children for accuracy. The breathing rate is noted. If children appear ill, blood pressure is measured. If children have a cough or breathing problems, a sensor is clipped on a finger or an earlobe to measure the oxygen concentration in blood (pulse oximetry).

As doctors examine children, they look for warning signs (such as an ill appearance, lethargy, listlessness, and inconsolability), noting particularly how children respond to being examined—for example, whether children are listless and passive or extremely irritable.

Occasionally, the fever itself can cause children to have some of the warning signs including lethargy, listlessness, fast breathing, and ill appearance. Doctors may give children fever-reducing medications (such as ibuprofen) and reevaluate them once the fever is reduced. It is reassuring when lethargic children become active and playful once the fever is reduced. On the other hand, it is worrisome when ill-appearing children remain ill-appearing despite a normal temperature.



Some Common Causes and Features of Fever in Children

Cause	Common Features*	Tests
Acute fever (lasting 14 days or less)		
Respiratory infections due to a virus	A runny or congested nose	
	Usually a sore throat and cough	A doctor's examination
	Sometimes swollen lymph nodes in the neck, without redness and tenderness	Sometimes testing of swab of nasal mucus
Other infections due to a virus	In some infants or children, no symptoms except fever	A doctor's examination
<u>Gastroenteritis</u>	Diarrhea	
	Often vomiting	
	Possibly recent contact with infected people or certain animals or consumption of contaminated food or water	A doctor's examination Sometimes examination and testing of stool
<u>Ear Infection (otitis media)</u>	Pain in one ear (difficult to detect in infants and young children who do not talk)	A doctor's examination
	Sometimes rubbing or pulling at the ear	
Throat infections (<u>pharyngitis</u>)	A red, swollen throat Pain when swallowing	A doctor's examination



MSD

Ear Infection (otitis

media)

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